Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam A · Form 2: Foundations

Multiple Choice Questions: Select the best answer. (1 point each)

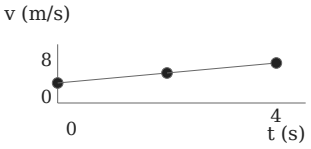
1. A velocity–time graph is a straight line from +4 m/s at t = 0 to +8 m/s at t = 4 s. What is the acceleration?

(A) 2 m/s²

(B) 0.5 m/s²

(C) 1 m/s²

(D) -1 m/s²



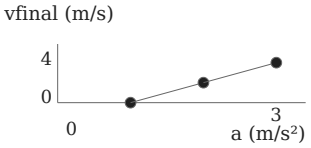
2. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1, 0); (2, 2); (3, 4). As a (m/s²) increases, what happens to vfinal (m/s)?

(A) increase

(B) no change

(C) decrease

(D) reversal midway



3. A ball is thrown vertically upward from the ground at 19.6 m/s. Ignore air resistance and use g = 9.8 m/s². Find the acceleration magnitude at the top.

(A) 9.8 m/s²

(B) -9.8 m/s²

(C) 19.6 m/s²

(D) 4.9 m/s²

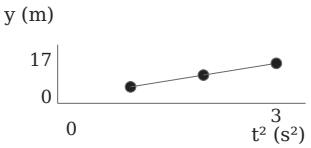
4. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². t² (s²) -> y (m): (1, 7); (2, 12); (3, 17). What is the mean of the first two vertical readings?

(A) 5.7

(B) 9.5

(C) 17.1

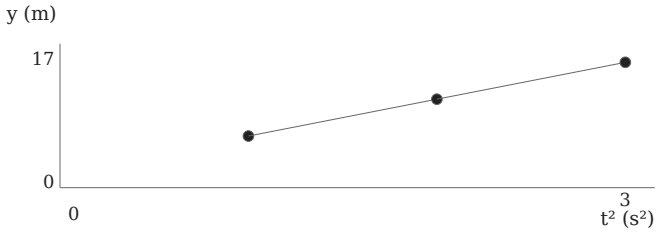
(D) 13.3



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². t² (s²) -> y (m): (1, 7); (2, 12); (3, 17).

a. For the drop sensor setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Plot the first and last measurements and describe the trend. **(3 pt)**

6. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.5, 7.5); (1, 6); (1.5, 4.5).

a. Build the model for runner separation from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1, 0); (2, 2); (3, 4). Build the model for velocity-area experiment from physics and predict both slope and intercept. **(3 pt)**